Example 4.14 In the Fig.4.38, $DE \parallel AC$ and $DC \parallel AP$. Prove that $\frac{BE}{EC} = \frac{BC}{CP}$.

Solution In ΔBPA, we have $DC \parallel AP$. By Basic Proportionality Theorem,

we have $\frac{BC}{CP} = \frac{BD}{DA}$ ...(1)

In ΔBCA, we have $DE \parallel AC$. By Basic Proportionality Theorem, we have,

$\frac{BE}{EC} = \frac{BD}{DA}$ ...(2)

From (1) and (2) we get, $\frac{BE}{EC} = \frac{BC}{CP}$. Hence proved.

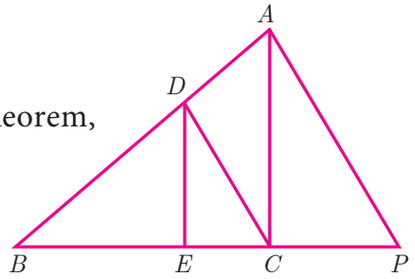


Fig. 4.38

Example 4.15 In the Fig.4.39, AD is the bisector of $\angle A$. If $BD = 4$ cm, $DC = 3$ cm and $AB = 6$ cm, find AC.

Solution In ΔABC, AD is the bisector of $\angle A$

By Angle Bisector Theorem

$$\frac{BD}{DC} = \frac{AB}{AC}$$

$\frac{4}{3} = \frac{6}{AC}$ gives $4AC = 18$. Hence, $AC = \frac{9}{2} = 4.5$ cm

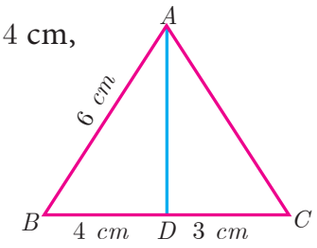


Fig. 4.39

Example 4.16 In the Fig. 4.40, AD is the bisector of $\angle BAC$, if $AB = 10$ cm, $AC = 14$ cm and BC = 6 cm. Find BD and DC.

Solution Let $BD = x$ cm, then $DC = (6-x)$cm

AD is the bisector of $\angle A$

By Angle Bisector Theorem

$$\frac{AB}{AC} = \frac{BD}{DC}$$

$\frac{10}{14} = \frac{x}{6-x}$ gives $\frac{5}{7} = \frac{x}{6-x}$

$12x = 30$ we get, $x = \frac{30}{12} = 2.5$ cm

Therefore, $BD = 2.5$ cm, $DC = 6 - x = 6 - 2.5 = 3.5$ cm

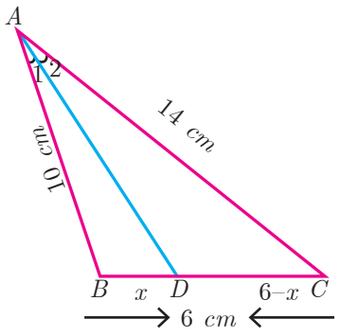


Fig. 4.40

Progress Check

1. A straight line drawn _____ to a side of a triangle divides the other two sides proportionally.
2. Basic Proportionality Theorem is also known as _____.

4 Geometry.indd 177 16-12-2020 01:55:01 PM